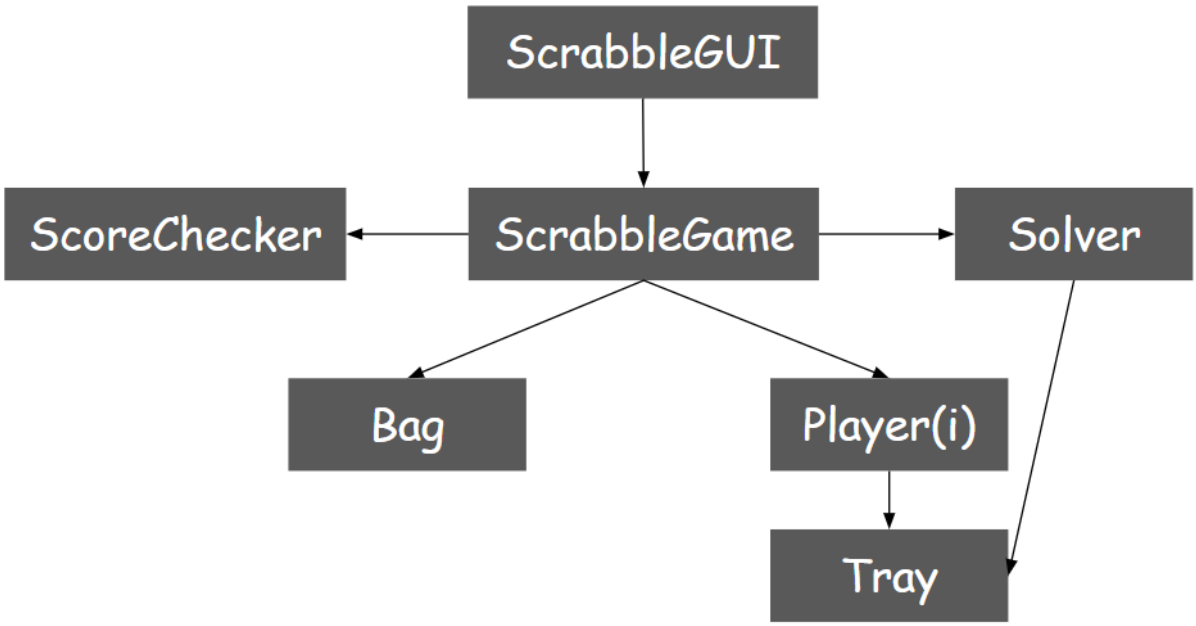
# Scrabble Design Diagram




## Description:

Tile – Represents a Scrabble Tile and stores its letter, value, and blank status. Used by most parts of the program.

Square – Represents one position on the board and stores multiplier information. Used by most parts of the program.

Dictionary – Stores valid words and valid prefixes used by the program. Used by most parts of the program.

Board – Represents the Scrabble board and stores all board squares. Used by most parts of the program.

ScoreChecker – Checks whether a play is legal and calculates its score.

ScrabbleGame – Controls the main Scrabble Game, including turns, scoring and game flow.

Solver – Finds the highest-scoring legal move for a given board and tray.

Player(i) – Represents the two players; the human player and the computer player.

Tray – Stores the tiles currently held by a player.

Bag – Stores the remaining Scrabble tiles and handles drawing and returning tiles.

ScrabbleGUI – Provides the JavaFX interface for playing the game.

Move – Stores a possible move, including the word, resulting board, and score.